Module 9 Lecture - Hypothesis Testing with One Sample

Introduction to Statistical Methods

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1 Overview and Introduction

1.1 Textbook Learning Objectives

- Differentiate between Type I and Type II Errors
- Describe hypothesis testing in general and in practice
- Conduct and interpret hypothesis tests for a single population mean, population standard deviation known.
- Conduct and interpret hypothesis tests for a single population mean, population standard deviation unknown.
- Conduct and interpret hypothesis tests for a single population proportion.

1.2 Instructor Learning Objectives

- Be able to identify different parts of a hypothesis given a narrative example
- Understand the connection between α, p-values, probability, and Type I and II errors
- Be able to work through practical inferential statistics computations for a single sample scenario

1.3 Introduction

- In statistics, hypothesis testing is the process by which we ________ whether data supports making a certain conclusion beyond a ________ doubt
- We have certain ________ to work through a hypothesis test:
  - Set up the ________ null and alternative hypotheses
  - Collect data in a ________
  - Determine an appropriate ________ test and distribution to represent our ________
  - Conclude whether we can ________ the null hypothesis or if we cannot

> **Important**
>
> I think it's important to emphasize that it is not our 'goal' to reject the null hypothesis - as empiricists, our orientation should be towards truth, not a certain outcome

2 Null and Alternative Hypotheses

2.1 Introduction

- A hypothesis is just a prediction or a statement of a possible ________
  - E.g., I *hypothesize (or predict)* that college students who are Gen Z have significantly worse attention than Millennial college students
  - Most scientific ________ begin with some hypothesis of what outcomes will look like

> Discuss: Consider that you are a middle-school teacher being asked by administration to implement an especially permissive attendance structure, write a hypothesis of what this will do to student math scores

- When we make ________ in statistics, we do so in a ________ pair by making a null hypothesis and an alternative hypothesis
  - In ________ statistical testing, we frame our inferential tests as helping us determine whether to reject the null - there is a whole other field of statistics called ________ statistics that has a somewhat different focus

> **Important**
>
> It's important to understand that, as part of the statistical testing process, we should always consider what our null and alternative hypothesis is!

2.2 Null Hypotheses

- A null hypothesis is a prediction or statement of ________ difference between two or more things
  - It is usually written as H_0
  - Opposite of alternative hypothesis
- A null hypothesis is often written to include an ________ sign such as with:
  - $=$

  - But also, $\geq$ & $\leq$
- Example: There is *no* difference in attention between Gen Z and Millennial college students
  - Represented in notation $H_0 : Attention_Z = Attention_{Millennial}$

Discuss: Write the null hypothesis for comparing mental health between women and men in the legal field, where they have equal levels of depression

2.3 Alternative Hypotheses

- Our alternative hypothesis suggests a difference between two or more things
  - It is most often represented as H_A or H_1
  - Opposite of Null Hypothesis
  - This is usually close to how we would phrase a ______ hypothesis
- The ______ hypothesis does *not* contain equal signs
  - This would include $\neq$, $>$, and $<$
- Example: There *is* a difference between Gen Z and Millennial college students in attention
  - Represented in notation $H_A : Attention_Z \neq Attention_{Millennial}$

Discuss: Write the alternative hypothesis for comparing mental health between women and men in the legal field, where women have higher levels of depression

3 Outcomes and the Type I and Type II Errors

3.1 Decision-making

- When we gather and ________ the data we gather from a sample, we use those results to make a decision
  - The decision comes down to whether we can ________ the null hypothesis or not (sometimes called 'retaining' the null hypothesis)
  - Ideally we avoid making a type I or type II ________ with our decision

Reality (Truth)	Decision: Reject H_0	Decision: Fail to Reject H_0
H_0 is True	**Type I Error (α)** False positive	**Correct Decision** (True negative)
H_1 is True	**Correct Decision** (True positive)	**Type II Error (β)** False negative

3.2 Types of Errors

- A type I error is when we ________ to reject the null hypothesis, when it is actually true
  - The ________ of a Type I error occurring (i.e., $P(TypeI)$) is given as α

Discuss: Review: where did we come across alpha before? Explain it's significance in the context of module 8!

- A type II error is when we decide to retain the null hypothesis when it is actually ________
  - The ________ of a Type II error occurring (i.e., $P(TypeII)$) is given as β

Important

When I was first trained on statistics, I was taught to view the Type I error as being more 'egregious' - 'the number one error to avoid!' - but realistically, they both result in flawed conclusions which could be dangerous

- Ideally, both α and β should be as ______ as possible, as to try an avoid making an error of any sort!
  - An extension of this is that power, the ______ we correctly reject the null when it is false
  - $Power = 1 - \beta$

? In a hypothetical experiment when comparing students before and after a new instructional style, I correctly decide I can reject the null hypothesis that there was no change. This is accurate, as there actually was a change. What error occurred here?

- (•) A) No
- ( ) B) error
- ( ) C) Type I
- ( ) D) Type II

Explanation:

______

4 Probability Distribution Needed for Hypothesis Testing

4.1 Introduction

- Its important that we ______ the sampling probability distribution we can use to represent a certain variable and it's mean or proportion
  - Effectively, this is where we are choosing a ______ - a very important step in our analysis!

Important

Choosing the 'right' test for certain situations is probably the most valuable skill a statistician has!

Discuss: Review: What is the formula for the SE of the mean and how does that tie back to the normal distribution of sample means?

Type of Hypothesis Test	Population Parameter	Estimated value (point estimate)	Probability Distribution Used
Hypothesis test for the mean, when the population standard deviation is known	Population mean μ	Sample mean $\bar{x}$	Normal distribution, $\overline{X} \sim N\left(\mu_X, \frac{\sigma_X}{\sqrt{n}}\right)$
Hypothesis test for the mean, when the population standard deviation is unknown and the distribution of the sample mean is approximately normal	Population mean μ	Sample mean $\bar{x}$	Student's t-distribution, t_{df}
Hypothesis test for proportions	Population proportion p	Sample proportion p'	Normal distribution, $P' \sim N\left(p, \sqrt{\frac{p \cdot q}{n}}\right)$

4.2 Assumptions

- An assumption is some prerequisite of the distribution we are using
- Take for example: the z-test for a population mean with the normal distribution. We need:
  - μ, σ, and the population must be normally distributed / large enough to appeal to the ________________
  - We must also have a ________________ random sample from the population

Discuss: Review: Describe why a simple random sample is often impractical. This is where research design intersects with analysis!

- The t-test for a population mean with the t-distribution assumes we also have a simple random sample from a normally distributed population

Important

Different inferential tests have different assumptions, and violating those assumptions can have big consequences! Too many researchers outright ignore these for convenience

5 Rare Events, the Sample, Decision, and Conclusion

5.1 Introduction

- While accounting for the ____________, statistics, parameters, and sampling distribution we have do help us make good choices about hypothesis testing - they don't tell us everything
  - We always have to remember that things in statistics are ____________!
  - With our sampling and data gathering we may run into a rare event
- To account for the ____________ of a rare event, we test the null hypothesis somehow

Discuss: Try re-explaining, in your own words, what 'probablistic' means and why it's readily applicable to the practice of statistics

5.2 Using the Sample to Test the Null Hypothesis

- This is where we introduce the p-value, which is the ____________ that, under the null hypothesis being ____________, our results will be as extreme as they are
  - Example: a test result returns a p-value of 0.07. Assuming the null hypothesis is true, this result only had a 7% chance of occurring.
- Effectively, when we ____________ against the null hypothesis and determine a p-value, we are trying to gauge how likely our results were to occur if the null hypothesis was true
  - If our results are especially ____________ under the null hypothesis (i.e., a low p-value), then we may be inclined to believe that our case is somehow truly different, and thus the null hypothesis is incorrect and can be ____________

Important

There are many misunderstandings people have about statistics, but failing to understand the meaning of p-values is probably the single most common and pervasive errors people make in interpreting statistics.

5.3 Decision and Conclusion

- How do we determine if the p-value is, "rare enough"?
  - Ideally, we test it against a preset/preconceived significance level, also given as α.
  - We ____________ whether our p-value is $\geq$ or $<$ our α

Important

This is the exact same alpha that showed up in regard to confidence level in the last module and earlier in our Type I error! We *choose* what we wish our significance level to be at the onset of our analyses.

- If you don't see further information, the most ____________ significance level is $\alpha = 0.05$
  - However, this isn't a hard set rule.

Discuss: Take a look at the prior discussion on Type I and Type II error - try using those terms to discuss why have too high of an alpha level might be risky

5.4 Tails of a Test

- A test may be described as two-tailed, left-tailed, or right-tailed, dependent on the ______ used in the alternative hypothesis
  - $H_a : P > 0.5 \rightarrow$ right-tailed (one-tailed)
  - $H_a : P < 0.5 \rightarrow$ left-tailed (one-tailed)
  - $H_a : P \neq 0.5 \rightarrow$ two-tailed
- This needs to be set as part of the study set up, not during analysis!

Discuss: Consider the following example: Johnny predicts that Samantha has more money than Becky right now. Is this a one-tailed or two-tailed test and why?

6 Conclusion

6.1 Recap

- Hypothesis testing is the backbone of inferential statistics, and testing against the null hypothesis is how we determine whether our results were just a chance "fluke" or likely some real difference!

- Proper hypothesis testing hinges upon wise identification of what values we have and what distribution best describes our variables. Each distribution and test we could use has a set of assumptions we need to be aware of
- P-values, alpha, beta, and the two hypotheses are all connected, and we must be careful to make the correct decisions - wary of the relative risk of Type I and Type II errors

Key Terms

A

alternative hypothesis A statement of non-equivalence between two or more things; opposite of the null hypothesis 4
assumption In statistics, an assumption is some key prerequisite or expectation in the distribution/test used 7

D

decision In statistics, this is our determination on whether we can reject the null hypothesis, based upon the gathered data and results of analyses upon it 5

H

hypothesis An empirical prediction that should be falsifiable. In statistics, it is a statement of comparison between two or more values 3
hypothesis testing A process for quantitatively testing predictions using inferential statistics, with the goal of inspecting whether we have sufficient evidence to reject the null hypothesis 2

N

null hypothesis A prediction or statement of no difference between two or more things, or non-relation; opposite of alternative hypothesis 3

P

p-value The probability, under the null hypothesis, that results will be as or more extreme than a given sample 9
power In statistics, power is the likelihood that we correctly reject the null hypothesis when it is false 6

R

rare event An occurrence of a unlikely outcome that appears to contradict our assumption or pre-existing belief (the null hypothesis) 8

S

significance level The preset alpha level we arbitrarily choose to test our hypotheses and p-values against 9

T

type I error When we make the decision to reject the null hypothesis, when it is actually true 5
type II error When we make the decision to retain or fail to reject the null hypothesis 5